Proposal: Foundational Model Development for Predicting Missing Ocean Color Data

Ocean color data, derived from satellite remote sensing, is multi- and hyperspectral, meaning it consists of light intensity at different color spectra. This type of data is critical for understanding marine ecosystems and biogeochemical properties. However, these datasets often have gaps caused by cloud cover and sensor issues. Image neural networks have the potential to improve on traditional gap-filling methods, like interpolation, by capturing the complex spatial, temporal, and spectral dynamics of ocean color patterns. We have developed a proof of concept using a U-Net CNN architecture (Figure 1) to gap-fill chlorophyll-a, which is a product developed from spectral data (Figure 2). https://agu.confex.com/agu/agu24/meetingapp.cgi/Paper/1700805

We propose the development of a foundational AI model[1] using self-supervised learning and the masking techniques developed in our U-Net application (Figure 3). Foundational AI models for earth science applications featured prominently in a number of sessions at the AGU Fall Conference (December 2024). During the hackweek, we would like to work with an engineer to develop a prototype foundational model. This foundational model will be designed to generalize across a variety of gap-filling applications, enabling reconstruction of missing data by leveraging the rich spectral and spatial context within hyperspectral ocean color rasters. Unlike task-specific supervised models, this approach learns directly from the inherent structure of the data, eliminating the need for fully observed training datasets.

Our proposed (or expected) methodology would involve training a masked autoencoder framework on global ocean color datasets. Synthetic gaps will be introduced during training to simulate real-world missing data scenarios using the approach we used for validating our U-Net model (Figure 3).. The encoder will capture latent representations of the visible data, while the decoder will reconstruct the full raster, including masked regions. We have a test dataset that we can use for a proof of concept for the development of a foundational mode (from our U-Net work)l.

Once developed, the idea is to test using the foundational model for diverse gap-filling applications with ocean color data. By creating a foundational model tailored for hyperspectral ocean color data, this project aims to create a valuable resource for advancing scientific research and operational monitoring but giving researchers a flexible tool for gap-filling ocean color data sets.

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[1]https://www.earthdata.nasa.gov/news/nasa-ibm-openly-release-geospatial-ai-foundation-model-nasa-earth-observation-data

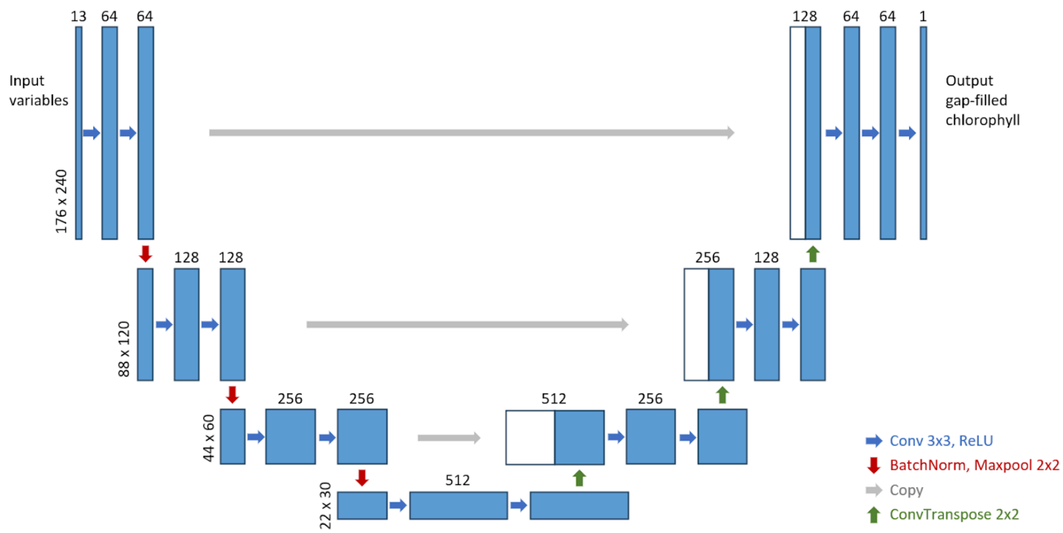

Figure 1. U-Net architecture for the gap-filling model (work by Yifei Hang, UW).

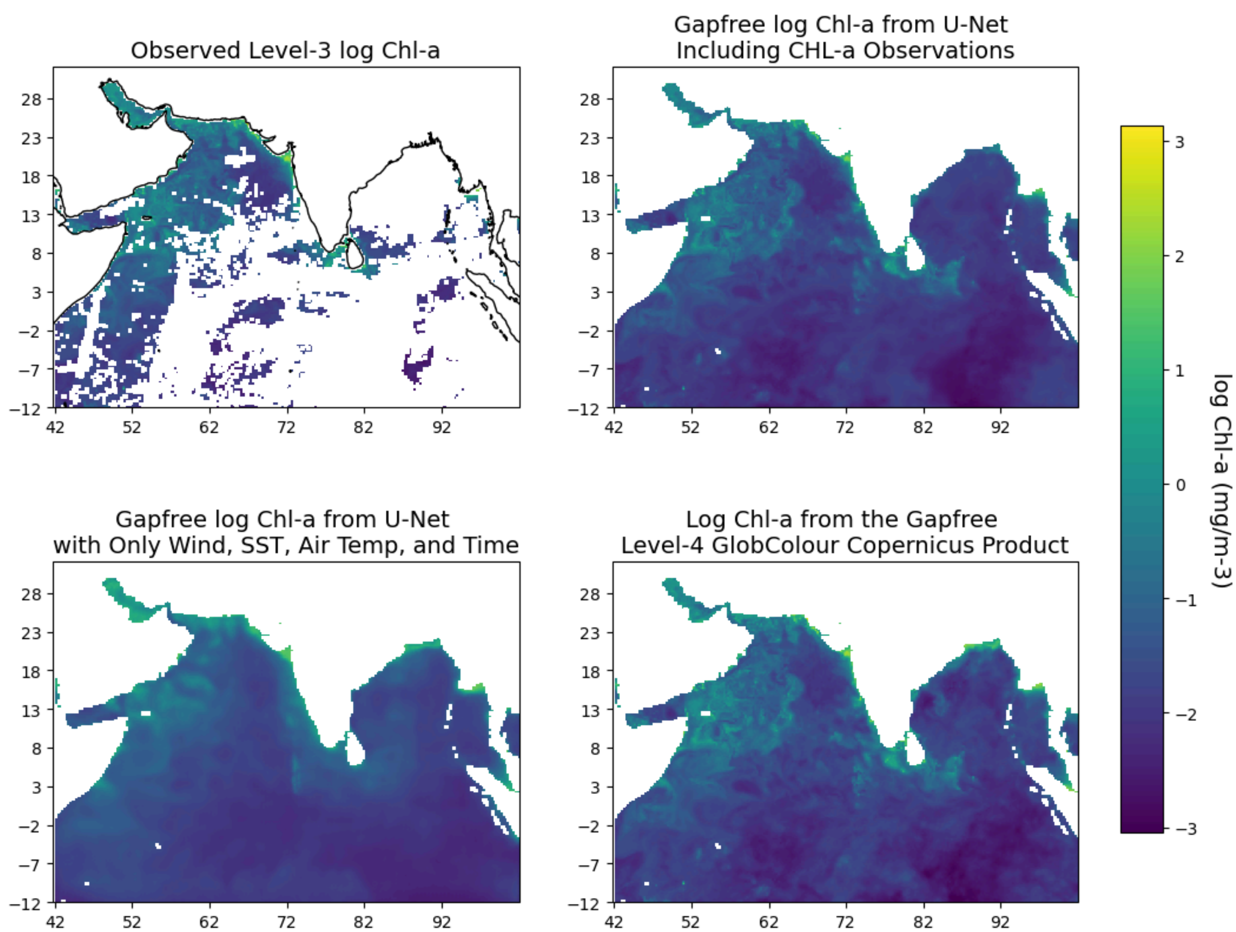

Figure 2. Example of gap-filling from U-Net CNNs versus the Copernicus GlobColour gap-filled product. Both are produced from the Level-3 data (top left). Work by Yifei Hang, UW and EE Holmes, NOAA.

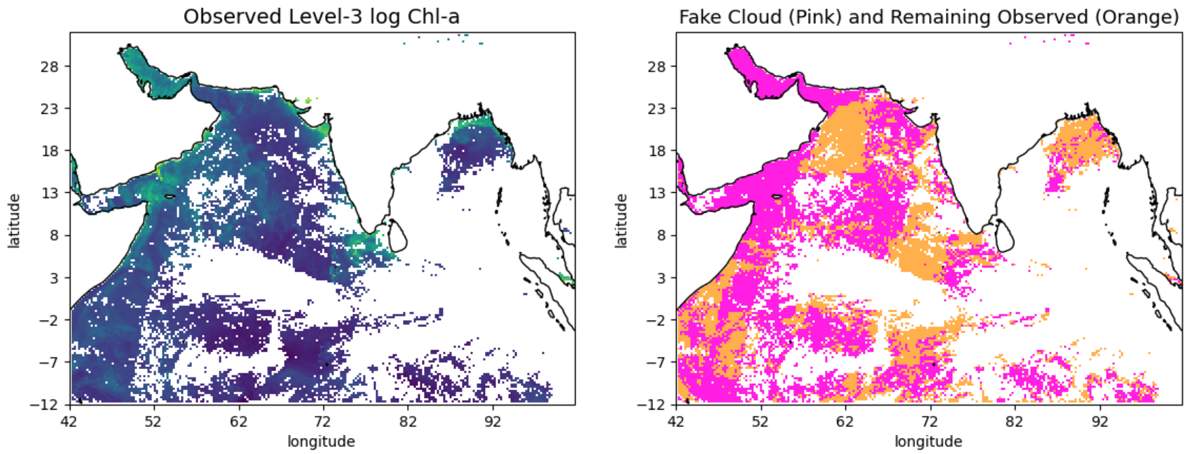

Figure 3: Application of masking using clouds 10 days in the future. This creates masks that are realistic in size and location. Work by Yifei Hang, UW and EE Holmes, NOAA.